

scripts/autonomous-qa/run-nav-challenges.sh

Boots ONE containerized-KVM emulator (via `lib-emulator.sh` — the same §6.AH path the autonomous-QA matrix uses) and runs the §6.AK cycle-coverage-map's supporting **navigation Challenges** (C24 / C46 / C55) against the already-built debug APK, then tears the emulator down.

Why it exists

The §6.AK cycle-coverage-map for a client cycle can list supporting nav Challenges (back-navigation, search-timeout interrupt, bottom-nav switching) alongside the keystone. `run-matrix.sh` only runs Challenge70; this script runs the *other* Challenges named in the coverage-map so each one gets a real per-Challenge device pass/fail in the §6.Z evidence.

Usage

```
# after the debug APK is built (app/build/outputs/apk/debug/app-  
debug.apk)  
scripts/autonomous-qa/run-nav-challenges.sh <evidence-dir> <fqcn>[  
    <fqcn>...]  
  
# example  
scripts/autonomous-qa/run-nav-challenges.sh \  
    .lava-ci-evidence/1079-nav \  
    lava.app.challenges.Challenge240nboardingBackNavigationTest \  
    lava.app.challenges.Challenge46SearchTimeoutInterruptTest \  
    lava.app.challenges.Challenge55MainScaffoldBottomNavSwitchesTest
```

Requires a pre-built `app-debug.apk` (exits 2 if missing — same pre-built-APK contract as `run-iteration.sh`). Installs `digital.vasic.lava.client.dev`.

Evidence

<evidence-dir>/ gets the JUnit XML + a per-test verdict (curated, tracked); <evidence-dir>/raw/ (gitignored) holds the logcat + screen recording.

Related

- `run-matrix.sh` / `run-iteration.sh` — the keystone (Challenge70) runner.
- `release-coldstart-canary.sh` — the release-variant R8 cold-start gate.
- Forensic anchor: `.lava-ci-evidence/sixth-law-incidents/2026-06-30-nav-challenges-first-device-run-brittleness.json`.